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Evaluation of inherent fructose, glucose and sucrose concentrations in tea leaves (*Camellia sinensis* L.) and in black tea

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ABSTRACT

Tea is the most popular beverage consumed in the world after water. In the present study, evaluation of inherent sugars viz. sucrose, glucose, and fructose in fresh tea leaves and in black tea were quantified using High Performance Liquid Chromatography. Fructose, glucose, and sucrose concentrations in the third and fourth tea leaves were significantly higher ($P \leq 0.05$) than that of the bud, first, and second leaves. The significantly lowest ($P \leq 0.05$) concentrations of glucose, sucrose, and fructose were observed in black tea manufactured using the bud and the first two leaves. There was no significant effect of time of plucking on glucose, sucrose, and fructose concentrations in fresh leaves and in black tea. There were significant ($P \leq 0.05$) changes in the concentrations of sucrose, glucose, and fructose that took place during the black tea manufacturing process. As sucrose and glucose have been allegedly used for the purpose of altering the appearance and heaviness of green and black teas, the results of this study would provide useful information for the tea industry in Sri Lanka and other countries to detect the adulterations of black and green teas with sucrose and glucose.

Introduction

Tea is the most popular beverage consumed in the world after water. All types of teas including black. green, oolong, white and pu-erh tea are mainly made from the buds and young leaves of tea plants, *Camellia sinensis* (L.) O. Kuntze belongs to the family Theaceae (Carvalho et al., 2014). The health benefits of tea consumption are reported to include antioxidant, anti-aging , anti-inflammatory, immuno-regulatory propertiesand reducing the risk of occurrence of heart disease, neurodegenerative diseases, diabetes, and cancer (Zhao et al., 2020a). Tea contains various bioactive compounds such as polyphenols, caffeine, saponins, theanine, γ-aminobutyric acid, vitamins, and triterpenoid saponins. Therefore, many research studies have indicated the potential health benefits of these biologically active molecules instead of primary metabolites, such as carbohydrates and sugars (Punyasiri et al., 2004; Carvalho et al., 2014; Zhao et al., 2020). However, over the past decade, it has been revealed that tea polysaccharide has many health benefits including antioxidant, antitumor, antibacterial and anti-aging properties, as well as the ability to reduce the risk of diabetes, improve immunity, and alleviate hepatotoxicity (Chen et al., 2008; Cai et al., 2013). It is stated that constituents other than biological active secondary metabolites, carbohydrate compounds that occurred in tea plants also play a vital role in the manufacture of high-quality tea, and it has been hypothesized that simple sugars play a part in the formation of tea aroma (Sanderson and Perera, 1966; Hara et al., 1995).

As carbohydrates contributed to the mellow taste of tea as well as exhibit some health benefits, a few studies have been carried out to quantify the carbohydrates including monosaccharides, disaccharides, and polysaccharides in tea leaves and in black tea (Wang et al., 2021). Previous research studies revealed that glucose and fructose content and sucrose content in tea shoot tips have been reported as 0.3–0.8% and 0.9–2.3% respectively while the total soluble carbohydrate content of tea shoot tips has been shown as 1.3–3.1% on a dry weight basis (Sanderson and Perera, 1965). In addition to that, in tea extract solids, around 4% polysaccharides, 0.15% pectin, and 6.5% sugars including fructose, glucose, sucrose, *m*-inositol along with small amounts of maltose and raffinose have been determined (Engelhardt, 2010).

As tea is solely a natural beverage derived from fresh tender leaves, the existence of sugar components in black tea is inevitable. However, in recent years, it is evident that further to satisfy the customer's sensory needs and earn unreasonable profits, low-cost sugar has been allegedly added to tea in the manufacturing process to improve the lustrousness and taste of tea and simultaneously achieve the purpose of weight gains (Wang et al., 2021). It is reported that some enterprises make illegal profits by adding sugar and glucose syrup to improve the taste, tighten the leaf strip and reduce costs in roasted green tea

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(Luqing et al., 2015). However, sucrose and glucose syrup added to tea has been hard to store because the growth of bacteria leads to form lumpy after moisture absorption and it may affect the quality of tea, increasing the risk of potential food safety issues and occur health risks (Luqing et al., 2015; Wang et al., 2021). As sucrose, glucose, and fructose are the main sugar components present in tea leaves, it is worth quantifying the inherent composition of these sugars in tea leaves and in black tea.

Although extensive research studies have been carried out on biochemical compounds in black tea, available information on its major sugars including sucrose, glucose, and fructose is scarce. Thus this study endeavored to determine three major soluble sugars, fructose, glucose, and sucrose in fresh tea leaves and in black tea quantitatively using High-Performance Liquid Chromatographic (HPLC) technique. In this study, the changes in sucrose, glucose, and fructose concentrations in tea leaves at different stages of leaf maturity, in different leaf standards, and time of plucking were studied. In addition to that, the effect of tea cultivar, climatic and seasonal variation on concentrations of sucrose, glucose, and fructose in fresh leaves and in black tea was investigated. Further, we evaluated the changes in sugar concentrations during the black tea manufacturing process. The findings from this study would provide useful for the tea industry in Sri Lanka and other countries to detect the adulterations of black and green teas.

2. Materials and methods

2.1. Materials and reagents

Zorbax NH2 column (150 × 4.6 mm (5μm)) was purchased from Agilent Technologies (Germany). Standard D-glucose (Purity - 99.5%), D-fructose (Purity - 99.99%), and D-sucrose (Purity - 99.99%) were purchased from Sigma-Aldrich (Germany). HPLC grade acetonitrile (Purity - 99.99%) was purchased from Sigma-Aldrich (Germany). Ultrapure water was purified using the Milli-Q system (Millipore, USA).

2.2. Extraction of the black tea/Freeze-dried tea sample

Black tea samples and freeze-dried fresh tea leaves (-40°C for 24 hrs, Freeze Dryer-Labconco) were ground through 0.5 mm mesh to homogenize the samples and a weight of 4 g of homogenized sample was extracted with 50 mL of distilled water at room temperature. Samples were brewed for 30 mins, in a magnetic shaker and 5 min in a centrifuged at 3500 r.p.m. (LAB-KITS). A volume of 1 mL of the extraction was diluted with 4 mL of 100% acetonitrile and filtered through 0.45 μM nylon filters (Agilent Technologies) prior to the HPLC injection (Jayawardhane et al., 2021). The dry matter content was determined by the loss in mass at 103°C in the oven (Memmert) for 6 hrs (ISO-1573, 1980).

2.3. HPLC instrumentation

HPLC instrumentation consisted of Agilent Technologies 1260 HPLC (Agilent Technologies, Germany). Separation module coupled to Refractive Index Detector (RID) (model G 7162A) system. The system was equipped with a built-in column oven and a vacuum degasser. Zorbax NH2 column (150 × 4.6 mm (5 μm)) was used for the analysis. Injection volume and flow rate were 10 μL and 1.5 mL min^{-1} respectively for all the standards and the samples. The method utilized an isocratic system with a mobile phase comprised of 83% acetonitrile: 17 % water. Column oven temperature is 33 ºC; Detector (RID) Temperature is 33 ºC. (Jayawardhane et al., 2021).

2.3.1. Experiment : A study on the effect of time of plucking on sucrose, glucose, and fructose concentrations in fresh tea leaves and in black tea

The first two leaves and a bud of tea cultivar DT 1 were plucked from 7 a.m. to 5 p.m. at two hourly intervals from St. Coombs Estate, Talawakelle in January 2021. Fresh tea leaves (100 g of each sample) were freeze-dried and from the same batch of fresh tea samples, a black tea sample (100 g of each sample) was miniature manufactured at the Biochemistry Division, Tea Research Institute of Sri Lanka. Sample collection was done in six different days.

(Miniature manufactured: ECM (Environmentally Control Miniature Manufacturing) System of Black Tea, Mini-manufacture (Teacraft). Conditions; Withering Difference between wet and dry bulb = 4°C, Relative Humidity = 75%, Rolling = Orthodox Roller, 15 min, Maceration = twice in roller; Sieving = Mesh No.8 (2.057 mm), Fermentation = ECM cupboard, dry bulb = 25°C, wet bulb = 25°C, Relative Humidity = 100 %, Drying = FBD (Fluid Bed Dryer), Inlet Temperature = 125°C, Outlet Temperature = 95°C,duration = 18-21 min, Removal of fiber = Fiber mat, PackagingTriple laminated aluminium foil).

2.3.2. Experiment : An investigation on changes in sucrose, glucose, and fructose concentrations in tea leaves at different stages of leaf maturity and in different leaf standards

The bud, first, second, third, and fourth leaves of tea cultivars, TRI 777, DN, DT 1, and TRI 2025 were collected separately, from St. Coombs Estate, Talawakelle, in December 2019 (Tea cultivars, TRI 777 and DT 1 are used to produce high-quality black tea while tea cultivars TRI 2025 and DN are used to produce moderate-quality black tea in Sri Lanka). Sample collection was done in three different days at the same time and the average values of three different days were considered. Freeze-dried fresh tea leaves (50 g of each sample) were used in this study.

The bud and first two leaves, bud, and first three leaves, bud and first four leaves and bud and first five leaves of tea cultivar (TRI 777) were collected separately, from St. Coombs Estate, Talawakelle, in December 2020. Samples collection was done in three different days. Black tea (100 g of each sample) was miniature-manufactured at the Biochemistry Division, Tea Research Institute of Sri Lanka.

2.3.3. Experiment : A study on the changes of sucrose, glucose, and fructose concentrations during black tea processing

Fresh tea leaves withered leaves, rolled leaves (1st, 2nd & 3rd), non-fermented, fermented, and black tea (1st, 2nd & 3rd) were collected from St. Coombs tea factory, Talawakelle, Sri Lanka. All the samples were dried at 60°C for 12 hours. The samples (100 g of each sample) were taken in five different days.

2.3.4. Experiment : A study on the effect of cultivar, climatic and seasonal variations (wet and dry seasons) on sucrose, glucose, and fructose concentrations in fresh leaves and in black tea

Fresh tea leaf samples (bud and the first two leaves) were collected from tea cultivars TRI 2023, TRI 2025, TRI 3055, TRI 4042, and TRI 4049, separately on the same day, at 10.00 a.m. in the morning in St. Joachim tea factory, Ratnapura (elevation: 29 m) and St. Coombs tea factory, Talawakelle (elevation: 1394 m), Sri Lanka. Freeze-dried fresh tea leaves (50 g of each sample) were used in this study.

The first two leaves and a bud were plucked from tea cultivars DT 1 and TRI 2025 in the second week of each month, at 10.00 am, and black tea (100 g of each sample) was miniature-manufactured at the Biochemistry Division, Tea Research Institute of Sri Lanka. Samples collected from May 2019 to December 2020, i.e. 18 months were used, excluding two months. (Average weather conditions; rainfall = 177.6 mm, minimum temperature = 14.2°C, maximum temperature = 23.1°C, and sunshine hours = 4.9, Soil type ;Talawakelle = Red yellow podzolic, Typic Hapludults, Soil type; Ratnapura = Red yellow podzolic, Typic Plinthudults).

Low grown black tea samples of three grades (Pekoe, BOPF, and BM) were collected from 15 tea factories during dry and wet seasons in 2018; altogether 90 black tea samples (250 g of each sample) were analyzed.

2.4. Sample preparation

The grounded fine powder samples (0.5 mm) from black tea, tea processing samples dried at 60°C, and freeze-dried fresh tea leaves were sealed in triple-laminated bags and stored in a cold room at 8 C° until use. All the samples were prepared following the procedure described above for HPLC analysis. Each extract was injected directly into the HPLC instrument. Glucose, sucrose, and fructose concentrations of all the samples were determined. All the concentration values are expressed as mg/g, dry weight basis.

2.5. Statistical analysis

One-way ANOVA was performed by using SAS version 9.1. Differences were considered significant at an alpha level of 0.05 at a 95% confidence level followed by Duncan's multiple range test as the mean separation technique.

Results and discussion

According to the results obtained, there was no significant effect of time of plucking on glucose, sucrose, and fructose concentrations in fresh leaves and in black tea. Sucrose and glucose concentrations in fresh leaves changed significantly ($P \leq 0.05$) during the black tea manufacturing process. The concentration of sucrose and glucose decreased significantly, whereas fructose concentration was not changed significantly during the tea manufacturing process. However, this study indicated that the concentration of sucrose and glucose in fresh tea leaves decreased significantly during the black tea manufacturing process. This may be due to the formation of the browning compounds/Maillard reaction products including acrylamide which occurred at temperatures above 120°C in the manufacturing of roasted green tea (Mizukami et al., 2006). The Maillard reaction is defined as a chemical reaction between reducing sugars (glucose and fructose) and amino acids which produces brown pigments with caramel flavor (Zheng et al., 2016). Amino acids, asparagine, aspartic acid, glutamic acid, and theanine in tea leaves are also statistically significant parameters for analyzing the acrylamide formation in roasted green tea products (Mizukami et al., 2006). Further, the amalgamation of carbohydrates with nitrogenous constituents in black tea leads to the augmentation of the natural color of oxidized polyphenolic compounds (Wickremasinghe, 1978). The quality of tea is generally reliant on the chemical composition of the green leaf used for its production and process parameters are carefully controlled during manufacture (Robertson, 1992).

Black tea is manufactured from the young tea shoots, perfectly comprising of the apical bud and the two youngest leaves. Bud and young leaves contain more tea polyphenols, caffeine, amino acids, and their concentration decrease significantly from the first leaf to the third leaf (Prawira-Atmaja et al., 2018). Young green tea shoots are extremely rich in polyphenolic compounds, the largest group being the catechins which constitute up to 30% of the dry weight of the material (Kumar et al., 2013). This may be due to, tea plants coordinating many metabolic pathways that simultaneously take place in young tea leaves in response to both developmental and environmental cues (Zhao et al., 2020b). Therefore, the processing of tea using old leaves affects the quality of black tea. Although black tea is manufactured from the young shoots (apical bud and the two youngest leaves), sometimes plucking involves the three/four leaves (Robertson, 1992; Prawira-Atmaja et al., 2018). Therefore, this study was undertaken to determine the variation of the glucose, sucrose, and fructose concentrations with the increasing maturity of tea leaves. The variation of sugar concentration of fructose, glucose, and sucrose in the development stage of tea leaves (bud, 1st, 2nd, 3rd, and 4th) was studied. This study revealed that there was a significant difference ($P \leq 0.05$) in fructose, glucose, and sucrose concentrations in different stages of leaf maturity, i.e. bud, 1st, 2nd, 3rd, and 4th leaves (see Table 1). Furthermore, there was no significant difference among the tea cultivars (TRI 777, DN, DT 1, and TRI 2025) used in this study.

Table 1
Concentrations of fructose, glucose, and sucrose in tea leaves at different stages of leaf maturity (n = 35).

Leaf maturity stage	Concentration (mg/g)		
	Fructose ±SD	Glucose ±SD	Sucrose ±SD
Bud	0.52^{b}±0.41	3.17^{b}±2.57	2.86^{c}±1.15
First leaf	1.65^{b}±1.57	2.65^{b}±0.96	5.52^{c}±2.35
Second leaf	1.93^{b}±1.38	4.82^{b}±3.79	11.28^{b}±6.85
Third leaf	4.53^{a}±2.33	10.76^{a}±6.77	25.73^{a}±4.52
Fourth leaf	5.17^{a}±1.78	12.14^{a}±2.67	29.13^{a}±3.34

Means with the same letter within a column are not significantly different.
n = Total number of samples, SD = Standard deviation.

Table 2
Fructose, glucose, and sucrose concentrations of black tea in different leaf standards (n = 22).

Leaf standard	Concentration (mg/g)		
	Fructose ±SD	Glucose ±SD	Sucrose ±SD
Bud & first 2 leaves	6.77^{c}±1.68	5.99^{c}±2.98	0.97^{c}±0.77
Bud & first 3 leaves	10.34^{b}±188	10.94^{b}±3.36	2.42^{c}±1.37
Bud & first 4 leaves	12.67^{a}±0.85	14.73^{a}±2.13	5.87^{b}±2.12
Bud & first 5 leaves	12.73^{a}±0.62	15.73^{a}±2.11	10.14^{a}±0.62

Means with the same letter within a column are not significantly different.
n = Total number of samples, SD = Standard deviation.

The findings showed that fructose and glucose concentrations in the third and fourth leaves were significantly ($P \leq 0.05$) higher than that of the bud, first, and second leaves. Sucrose concentration in the third and fourth leaves was significantly higher ($P \leq 0.05$) than that of the second leaf. Moreover, a significantly ($P \leq 0.05$) lower content of sucrose was reported in the bud and firstleaf. It is reported that the concentration of chlorophyll increased significantly with the development stage of the leaf while total polyphenol tends to decrease with leaf maturity. Pearson Correlation analysis showed that chlorophyll content was negatively correlated ($r = -0.83$; $p = 0.05$) with total polyphenol during the developmental stage of tea leaves (Prawira-Atmaja et al., 2018). The results of this study clearly showed that the concentration of sugar components, glucose, sucrose, and fructose have increased with the maturity of tea leaves.

In most black tea manufacturing countries, the recommended plucking standard for tea manufacturing is tender tea shoots of two leaves and a bud (Wickremasinghe, 1978). However, occasionally coarse plucking such as bud and the first three leaves, bud and the first four leaves, and bud and the first five leaves is carried out to increase extra biomass achieved in a plucking round (Owuor et al., 1987). Accordingly, processing of tea using old leaves (third and fourth leaves) may have the potential to increase the concentrations of glucose, sucrose, and fructose in black tea. Consequently, an investigation on changes in sugar concentration in different leaf standards for black tea was carried out. The leaf standards shown in Table 2 were used in this study.

The findings from this study showed that the significantly ($P \leq 0.05$) lowest glucose, sucrose, and fructose concentrations were found in black tea manufactured using the bud and the first two leaves. And also, the significantly high glucose and fructose concentration was found in black tea manufacturing using the bud and first three leaves, and bud and first four leaves while the significantly ($P \leq 0.05$) highest content of sucrose was reported in black tea manufactured using the bud and fifth leaves.

Black tea is manufactured from young shoots (apical bud and the two youngest leaves). In practice, even though much plucking is car-

Table 3
Concentrations of fructose, glucose, and sucrose in tea leaves at different manufacturing stages (n = 84).

Black teamanufacturing stages	Concentration (mg/g)		
	Fructose±SD	Glucose±SD	Sucrose±SD
Fresh leaves	11.10^{c}±2.45	15.09^{c}±2.94	10.08^{b}±2.75
Withered leaves	10.86^{c}±1.41	14.09^{d}±1.16	13.18^{a}±1.61
Rolled leaves (1st)	18.48^{a}±2.78	20.31^{a,b}±2.73	5.45^{b,c}±2.85
Non fermented Dhool (1st)	19.04^{a}±2.22	21.14^{a}±2.61	6.08^{b,c}±2.26
Rolled leaves (2nd)	18.47^{a}±2.59	20.45^{a,b}±2.57	6.51^{b,c}±1.91
Non fermented Dhool (2nd)	18.19^{a,b}±2.48	20.35^{a,b}±2.79	6.41^{b,c}±2.96
Rolled leaves (3rd)	17.63^{a,b}±2.78	19.00^{a,b,c}±2.98	7.39^{b,c}±2.58
Non fermented Dhool (3rd)	17.37^{a,b}±2.96	19.18^{a,b,c}±2.75	7.33^{b,c}±2.67
Fermented Dhool (1st)	17.80$^{a.b}$±1.92	19.84^{b,a}±1.82	6.92^{b,c}±2.80
Fermented Dhool (2nd)	18.91^{a}±2.20	20.58^{a,b}±2.68	4.89^{c}±2.99
Black Tea (1st Dhool)	14.73^{c} ±1.18	16.42^{b,c,d}±1.92	12.71^{a}±4.74
Black Tea (2nd Dhool)	15.86^{a,b}±3.91	17.00^{a,b,c,d}±4.84	12.27^{a}±2.71
Black Tea (3rd Dhool)	14.00^{c}±2.51	15.08^{c,d}±3.39	13.62^{a}±4.49

Means with the same letter within a column are not significantly different.
n = Total number of samples, SD = Standard deviation.

ried out by hand, the composition of the plucked green leaf could be two, three, and four and a bud resulting in the manufacture of widely varying shoot sizes and ages (Robertson, 1992). And also, catechin constituents and total polyphenol content change significantly with leaf age (Samanta et al., 2017). Leaf standards affect the chemical composition and quality of tea, as stalks have a different chemical composition compared to leaves (Engelhardt, 2010). These changes have a significant effect on the products of fermentation and black tea quality and are a major reason for the preferred practice of plucking young flush (Robertson, 1992; Engelhardt, 2010). Further, the findings from this study clearly revealed that sucrose, glucose, and fructose concentrations in tea flush were increased with the plucking of mature leaves i.e. third, fourth, and fifth leaves. Accordingly, the leaf standard of green leaves is one of the most crucial factors, evidently associated with sugar levels in black tea.

Tea manufacturing is solely a mechanically controlled biological process which, ultimately ends up with a chemical composition derived through the natural transformation of chemical constituents found in fresh leaves (Robertson, 1992; Engelhardt, 2010). According to the previous results in this study, sucrose, glucose, and fructose concentrations in fresh leaves have changed significantly ($P \leq 0.05$) during the black tea manufacturing process. Accordingly, this study was carried out to determine changes in sugar concentration in different tea processing stages. Although, the typical drying temperature is around 90–95°C for 21 minutes using for the black tea manufacturing process (Samanta et al., 2015), in this study, samples collected from the different manufacturing stages were dried at 60 °C. According to the results, it was found that significant ($P \leq 0.05$) changes in the levels of sucrose, glucose, and fructose took place during the black tea manufacturing process. Further, glucose, sucrose, and fructose concentrations have fluctuated drastically during the tea manufacturing process. This may be due to intense chemical and biochemical changes occurring during each of these stages of the black tea manufacturing process (Robertson, 1992). According to the results obtained (see Table 3), the highest fructose concentration was observed in non-fermented dhool (1st) and rolled tea leaves (2nd) whereas the lowest fructose concentration was shown in the withered leaves. The highest glucose concentration was observed in non-fermented dhool (1st) whereas lowest glucose concentration showed in the withered leaves. The significantly highest sucrose concentration was observed in black tea and the lowest sucrose concentration was found in fermented dhool. The set of smaller tea particles separated from the bulk after maceration in the first rotorvane is called the “first dhool” and spread directly on the floor or table for the aeration process. The oversize particles after the separation process undergo maceration in rotorvane and a separation process again. The second set of separated smaller tea particles is called the “second dhool” and the maceration and separation process are repeated again twice more to produce the “third dhool” and the "fourth dhool". All the dhools collected from the separation process are spread on the floor or table directly(Keegel, 1983). A previous study reported that remarkable changes in the levels of the various carbohydrate fractions have occurred during tea processing and soluble carbohydrates are increased during tea processing (Sanderson and Perera, 1965).

Previous studies revealed that inherent glucose, sucrose, and fructose concentrations in processed tea and in fresh tea leaves are varied. The range of sucrose, glucose, and fructose concentrations of tea leaves have been determined as 10.49-55.23, 0.81-17.22, and 1.05-13.62 mg/g respectively in 82 tea cultivars (Mizukami et al., 2006). In addition, it is revealed that Longjing and Bi Luo Chun tea, two different green tea types contain 0.47-0.86, 0.38- 0.79 mg/g of fructose, 0.59-0.98, 0.73-1.31 mg/g of glucose and 8.20-9.94, 6.16-7.81 mg/g of sucrose contents respectively (Ding et al., 2002). Additionally, it is reported that instant black tea contains 69.2-76.2 mg/g, 33.0-41.4 mg/g, and 9.5-24.8 mg/g of fructose, glucose, and sucrose respectively (Alasalvar et al., 2013). Green tea contains various kinds of chemical substances contributing to the taste, the sweetness of sugars which have been contained in green tea might be required for the good taste of green tea (Ding et al., 2002; Liu et al., 2018). The flavor of oolong tea has been reported to be related to a combination of various compounds including soluble sugar (Liu et al., 2018). Accordingly, sugars also play a significant role in the enhancement of the quality of processed tea.

The investigation of the effect of clonal and climatic variations on inherent glucose, sucrose, and fructose content in tea leaves was undertaken. According to the results, the inherent glucose, sucrose, and fructose sugar concentration of the selected tea cultivars was not significantly varied. Furthermore, there was no significant interaction between tea cultivars and the regions. Further, sucrose and fructose concentrations in tea cultivars were not significantly different between the regions. There was a significant difference ($P \leq 0.05$) in glucose concentration, between the regions, up and low country. The inherent glucose concentration in tea cultivars in up country was higher than that of low country in Sri Lanka.

For the same purpose, another study was conducted extending the time period to determine the effect of climatic and clonal variation on these sugar components. According to the results obtained, for all three sugar components, there was no significant difference between tea cultivars DT 1 and TRI 2025. Spearman Correlation Analysis was performed to determine the correlation of weather data including total rainfall, minimum and maximum temperatures, and sunshine hours with the inherent glucose, sucrose, and fructose concentrations. According to the statistical results, there was no significant correlation between weather data and glucose, sucrose, and fructose concentrations in tea cultivars used.

Seasonal variations may have the potential to affect the inherent glucose, fructose, and sucrose concentrations in tea leaves. Thus, this study was carried out to determine the effects of seasonal variations (wet and dry seasons) and black tea grades on the inherent glucose, fructose, and sucrose concentrations in black tea.

A seasonal variation in sugar levels in black tea is evident from the data shown in Table 4. These results revealed that there was a significant difference ($P \leq 0.05$) in fructose and sucrose concentrations in black tea between dry and wet seasons. The inherent sugar levels of fructose and sucrose in black tea in the dry season were higher than that in the wet season. It is reported that an increase in the concentration of soluble sugars such as glucose, sucrose, and fructose enhances plant tolerance to several abiotic stresses, such as drought, salinity, and cold (Rosa et al., 2009; Khaleghi et al., 2019). Meanwhile, the existence of a dry spell in the environment at the time of plucking also becomes a causal factor in

Table 4
Fructose, glucose, and sucrose concentrations in black tea during the wet and dry seasons (n = 90).

Season	Concentration (mg/g)					
	Fructose		Glucose		Sucrose	
	Range	Mean±SD	Range	Mean±SD	Range	Mean±SD
Dry Season	2.89 - 9.33	6.45[a]±1.6	2.04 – 12.22	6.70[a]±2.15	1.82 – 9.40	4.69 [a]±1.59
Wet Season	2.46 - 7.49	5.34[b]±1.53	1.63 – 9.22	6.31[a.]±1.91	0.72 – 5.76	2.95[b]±1.49

Means with the same letter within a column are not significantly different.
n = Total number of samples, SD = Standard deviation.

Table 5
Fructose, glucose, and sucrose concentrationsin different black tea grades (n = 90).

Black tea grade	Concentration (mg/g)		
	Fructose±SD	Glucose±SD	Sucrose±SD
Pekoe	5.86[a]±1.75	6.33[a]±2.42	3.07[b]±1.54
BOPF	5.81[a]±1.73	6.24[a]±1.96	3.73[b]±1.44
BM	6.03[a]±1.59	6.95[a]±2.06	4.67[a]±1.99

Means with the same letter within a column are not significantly different.
n = Total number of samples, SD = Standard deviation.

this regard as in terms of plant physiological responses, plants tend to synthesize more soluble sugar components in leaves to gain an osmotic potential under the dry spell periods (Rosa et al., 2009; Khaleghi et al., 2019). Further, it is reported that sucrose regulates polyphenol biosynthesis in the tea plants and promotes the transport of polyphenols, and changes the aroma composition of the tea plants (Qian et al., 2018; Liu et al., 2018).

Tea grading is a process that involves the separation of fired tea into various shapes and sizes confirming trade requirements and making all the difference to the value of tea (Wickremasinghe, 1978; Keegel, 1983). There is no strict definition for tea grading and this grade specification is entirely artificial and varies from country to country, in addition to the size and shapes of the particles, the distribution of different portions of the fresh leaves is also represented by different black tea grades (Wickremasinghe, 1978; Keegel, 1983). Therefore, it may have the potential to change these sugar components in different types of tea grades. Accordingly, tea grades including BM, Pekoe, and BOPF were used in this study. Tea grade, Pekoe is the largest of the leafy grades, which derives from thriving withered leaves. BOPF is relatively a smaller-sized grade, taken through No 14, 16, or 18 meshes, and derives from the finer leaf, and BM grade is an off-grade (Keegel, 1983).

The findings demonstrated that there was no significant difference in fructose and glucose concentrations among black tea grades used. However, there was a significant difference ($P \leq 0.05$) in sucrose concentration in black tea grades. The highest sucrose concentration was found in BM grade whereas the lowest concentration was in the Pekoe and BOPF grades (see Table 5). However, in this study, it is evident that high concentrations of sugar components were found in mature tea leaves compared to young tea leaves. As tea grade, BM is an off-grade; it may have the potential to contain a high level of sucrose concentration in BM grade, compared to Pekoe and BOPF grades. BM grade is a mixture of flaky tea and it consists of flat leafy and stalks while BOPF is a smaller-sized grade and it should be derived from the finer leaf (Keegel, 1983). However, Pekoe is the largest of the leafy grades and it is inclined to contain a considerable quantity of cut leaf (Keegel, 1983). Furthermore, the shape and size of the particles and the distribution of different portions such as tip, stalk, etc. of the fresh leaves which represent “different black tea grades” may have had a significant influence on the concentrations of sucrose component in black tea.

The results of this study would be helpful to get a comprehensive understanding of how inherent glucose, fructose, and sucrose concentrations were varied in fresh tea leaves, in different tea processing stages, and in black tea. And also, the findings of this study would provide a positive impact for the tea promotion, as the tendency for the sugar-free beverages is being prevailed worldwide and low content of sucrose, glucose, and fructose was reported in black tea. In addition to that, whether the tea is drunk without sugar and milk or other additives, the calorific value of the tea brew is almost zero compared to other drinks and beverages that generally have constituents that may be rich in calories (Modder and Amarakoon, 2002). The findings from this study are in line with this statement since black tea contains relatively low inherent glucose, sucrose, and fructose concentrations compared to other beverages.

4. Conclusion

In this study, the evaluation of sucrose, glucose, and fructose concentrations in fresh tea leaves and in black tea was investigated. The findings indicated that the concentration of sugar components, glucose, sucrose, and fructose have been increased with the leaf maturity and poor leaf standards. Time of plucking was not significantly affected the concentrations of glucose, sucrose, and fructose in tea leaves and in black tea. Sucrose, glucose, and fructose concentrations in tea leaves changed significantly during the black tea manufacturing process. As tea manufacturers and consumers have been concerned about tea authentication and identification, the results obtained in this study would be helpful to identify black and green teas adulterated with glucoseand sucrose.

Ethical Statement

This is to inform you that in this study, we have not been involved in any animal and human studies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Glucose, sucrose and fructose concentrations in fresh leaves and black tea (n = 66)

	Concentration (mg/g)		
	Fructose±SD	Glucose±SD	Sucrose±SD
Fresh leaves	3.5^{a}±0.69	8.40^{a}±1.51	9.11^{a}±6.39
Black Tea	3.5^{a}±1.69	2.99^{b}±2.66	4.53^{b}±2.38

Means with the same letter within a column are not significantly different. n = Total number of samples, SD = Standard deviation

Glucose, sucrose and fructose concentrations in fresh leaves at different regions (n = 20)

	Concentration (mg/g)		
Region	Fructose ±SD	Glucose ±SD	Sucrose ±SD
Low country	4.65^{a}±2.24	6.14^{b}±3.23	18.62^{a}±12.28
Up country	5.96^{a}±1.6	9.84^{a}±3.57	15.74^{a}±10.41

Means with the same letter within a column are not significantly different n = Total number of samples, SD = Standard deviation

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